Shu Pu

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Education

University of California, Santa Cruz Sep 2026 – Jun 2031 (Expected)
Ph.D. in Computer Science and Engineering

Selected Publications

- ***i1:*** *A Simple and Fully Open Recipe for Strong Text-to-Image Models* [PDF]
  Boya Zeng, Tianze Luo, **Shu Pu**, Jucheng Shen, Taiming Lu, Gabriel Sarch, Zhuang Liu.
- ***Memorization in 3D Shape Generation:*** *An Empirical Study* [PDF]
  Shu Pu, Boya Zeng, Kaichen Zhou, Mengyu Wang, Zhuang Liu.
  CVPR'26 Findings.
- ***Judge Anything:*** *MLLM as a Judge Across Any Modality* [PDF]
  Shu Pu*, Yaochen Wang*, Dongping Chen, Yuhang Chen, Guohao Wang, Qi Qin, Zhongyi Zhang, Zhiyuan Zhang, Zetong Zhou, Shuang Gong, Yi Gui, Yao Wan, Philip S. Yu.
  KDD'25 DB Track (Oral).
- ***ISG-Bench:*** *Interleaved Scene Graph for Text-to-Image Generation Assessment* [PDF]
  Dongping Chen*, Ruoxi Chen*, **Shu Pu***, Zhaoyi Liu*, Yanru Wu*, Caixi Chen*, Benlin Liu*, Yue Huang, Yao Wan, Pan Zhou, Ranjay Krishna.
  ICLR'25 (Spotlight 5.1%).
- ***Thinking Before Looking:*** *Improving Multimodal LLM Reasoning via Mitigating Visual Hallucination* [PDF]
  Haojie Zheng*, Tianyang Xu*, Hanchi Sun, **Shu Pu**, Ruoxi Chen, Lichao Sun.

* Equal contribution, where indicated.

Experience

Ph.D. Student, University of California, Santa Cruz Sep 2026 – Present
Advisors: Cihang Xie and Yuyin Zhou

- Investigating out-of-sight dynamics for consistent, self-evolving 4D world generation conditioned on text, context frames, and camera trajectories.
- Deploying video generation models on TPUs with JAX.

Research Intern, Princeton University Mar 2025 – May 2026
Advisor: Zhuang Liu

- Led research on memorization in 3D generative models, developing an evaluation framework and conducting controlled experiments on data and modeling factors.
- Contributed to TPU-based training, model evaluation, and codebase development for **i1**, an open-source text-to-image generation model.

Research Intern, Huazhong University of Science and Technology Jun 2024 – Mar 2025
Advisor: Yao Wan

- Co-led **ISG-Bench**, driving method design, benchmark construction, core implementation, and paper writing; independently developed **ISG-Agent**.
- Led **Judge Anything**, developing a human-aligned multimodal generation evaluation benchmark and *Omni-Arena*, supported by fine-grained human analysis.

Research Intern, Lehigh University (Part-time) Jun 2024 – Aug 2024
Advisor: Lichao Sun

- Contributed to **Visual Inference Chain**, a multimodal LLM framework for mitigating visual hallucinations.

Skills & Interests

Programming: Python, C++
ML Frameworks: PyTorch, JAX
Research: Generative Models, Diffusion Models, Multimodal Learning, 3D/4D Generation
Computing: Model training, fine-tuning, inference, and evaluation; basic familiarity with CUDA, XLA, and GPU/TPU architectures.
Research Interests: Visual Generation, 3D/4D World Modeling, Multimodal Learning, Spatial Intelligence.
Hobbies: Football, Swimming, R&B.